Section A Data fundamentals

① A database can be defined as a place where information is stored in an organised way so that it is easy to find and make use of.

E.g. Banks store people's account details in a database

* Apps/systems like dropbox or google stores data in in a database

② Data~~b~~ Warehouse - A datawarehouse stores a lot of data from different places over a long period of time. This is used in the study of data and helps companies/businesses make decisi

* A database is used to store data securely, while a datawarehou stores data for analytical processing

③ Datalake - A datalake stores data in its original form, even if it is structured or unstructured.

* The difference between a datalake is that it stores raw data that has not been processed yet, whereas a data warehouse stores data that has been filtered and is organized.

* A business would use a data lake when they want to store large amounts of data and decide later what to do with that data

④ Datamart - This is a smaller part of the data warehouse will focus on a certain department, like ~~engine~~ HR or marketing.

- It is used internally when team members choose to access that they need quickly

⑤ Types of data
Structured & unstructured
- Structured data is organised and it is ~~easy~~ stored in tables, e.g spreadsheets.
- Semi structured data has some organisation to it, however, i is not fully structured e.g email
- Unstructured data has no deal to it. e.g songs, pictures.

⑥ Data modeling - This is the process of planning how data wi be stored and organised in a system.
* It is important because it helps to ensure that data the been stored is easy to use and understand.
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⑦ Data mining - This is the process of analysing data f patterns and useful information.
E.g store use data mining to understand consumer habits while institutions like SARS or the bank use it to fraud.

⑧ DBM - Database management system is a software that users to be interact with the ~~data~~ database
Oracle - for large businesses
GCP - google cloud platform for users
~~Microsoft~~ PostgreSQL - for advanced tasks.
- MySQL - for websites

(i) ETL - Extract, Transform & Load

- Extract means to collect data from a source/sources that vary
- Transform means filtering, cleaning and organising data
- Loading means storing the data in a system

This process is important because it helps to make sure that data is accurate and ready for analysis

(ii) Data Storage formats

CVS (This is the most basic. Data is stored as plain text, with values separated by commas) Easy to read and works with almost everything

JSON (JavaScript Object Notation) Stores data in a nested, key value format. It is flexible and works with semi-structured data But, it can get heavy or slow when the file gets too big

Parquet (A column based storage format commonly ~~s~~used in big tools) It is efficient because it stores data ~~in~~ by columns instead of rows making storage space less.

Avro (A row based format that ~~use~~ stores data along with it's schema. It is useful for data streaming.

(iii) Sector B

(iv) AI - Normally known as Artificial Intelligence.

This is the ability of machines to perform tasks that normally require human ~~the~~ intelligence.

Narrow AI (Weak AI) is designed to perform specific within a limited context, while General AI (Strong A a theoretical human-level Intelligence that can learn, re apply knowledge ~~consider~~

(12) ML (Machine Learning) allows systems to learn from data instead of being programmed directly.

Supervised Learning - Trains models using data sets, where sets are paired with known outputs

Unsupervised Learning - Uses data that is unlabeled to discover hidden patterns or relations within the data autonomously.

Reinforcement Learning - An agent learns to makes decisions through trial & error by interacting with an environment to maximise cumulative rewards.

(13) Deep Learning

A type of machine learning that uses neural networks to process complex data

Neural networks: Systems that mimic the human brain to recognize patterns

(14) NLP - Natural Language Processing helps machines to understand human language

e.g Chatbot, voice assistants apps.

(15) LLM

A large language model that is AI that is trained to understand text data and generate language e.g Chat gp

(16) This is AI that creates text, images or music e.g Gemini AI, Chat GPT

(17) Training dataset

This is data used to train / teach an AI model

This is impotante because it will detemine the kind of result that an AI tool will yield. Poor or biased to inaccurate results.

(20) Computer vision - This allows machines to understand images
e.g face recognition

Section C: APIs & Tools

(21) An API allows different software systems to communicate
An API acts like a middleman who delivers requests and responses e.g a waiter taking your order to the kitchen
* They are important because they connect different systems

(22) API key is a unique code that ~~client~~ is used to access an API (Application program interface). If exposed there could be potential security breaches or data stolen

(23) REST : Representational State transfer
Methods
Get - Retrieve data
Post - Send data
Put - Update data
Delete - Remove data

(24) JSON - Java Script Object Notation : This is a text based interchange format used to store and transit structured data
e.g Name : Jabulani Age : 26

(25) Prompt Engineering - This means giving clear instruction in order to generate something
Bad prompt: "Write me something"
Better prompt: "Write me a short story about Cinderella

(24) This applies the same action to every item on a list
e.g doubling all the numbers on a list.

(25) This means using services like storage over the internet
Iaas gives infrastructure
Paas gives tools
Saas gives software like gmail

(26) ~~This~~ Version control helps to track changes in code over time. Git is commonly used for this.
- A commit is saving changes, and a branch is making an separate versions of a project

(27) A Jupyter notebook is a tool used to run and write codes interactively
- It is mostly used with python and is popular because it allows users to see results immediately.

(28) A Python
- This is a programming language that is widely used data analytics because it is easily learnable and pow
Some libraries include pandas, ~~m~~numpy and scikit -

Section D

(29) Data privacy is about protecting personal information from being misused
- GDPR is a law in Europe that protects people's data and gives them control over how it is use

(30) Data ethics is about using data in a responsi fair way. e.g using data without permission, c

biased systems or misleading people with data.

(31) Data governance is the process of managing data properly to ensure that it is accurate, secure and accessible.

(32) Bias AI ~~is~~ happens when a system produces unfair results because of data it was trained on.

33 Roles comparison

- A data analyst focuses on analysing data and creating reports on their analysis.
- A data scientist builds models, ~~and~~ cleans, collects and analyses large complex datasets to build models using machine learning
- Data engineer designs, builds and maintains data pipelines and infrastructure.
- AI engineer builds AI systems

(34) Business intelligence involves using data to make better decision. This includes using tools like power BI, ~~tab~~ and tableau to help businesses understand their data through visual reports

(35) I found that AI and generative AI are interesting because ~~they~~ this shows how technology can have the impact of creating content that is similar to humans

(36) Data modeling was the most difficult for me to use at first, but used online resources to understand

(37) I believe data and AI play a role in my ~~ca~~ career especially marketing because it will help me understand my audience better. Create more engaging content.

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